Exponential sample lower bounds for Bayesian Gaussian process classification

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Abstract

We prove an exponential sample lower bound for the exact Gaussian sign posterior associated with saturated neural networks on the Boolean cube. At every fixed depth, there are constants $c, \varepsilon > 0$ such that, for every sample size $n \le e^{cd}$, independent uniform examples of d-bit parity leave a fraction at least ε of inputs without correct-label probability greater than $2/3$, with probability at least $3/4$ over training inputs. The result applies to the width limit of Bayesian tanh networks conditioned before training to have a fixed absolute margin at every input, for every fixed exponent $\gamma < 0$. The proof combines sharp Fourier level masses, random tail-covariance bounds, and a decomposition into low degrees, a thin middle band, and a high-degree remainder. A fixed-threshold Lipschitz sampling argument controls the negative mass of the actual posterior mean and closes the gaps between successive Fourier levels. No monotonicity in sample size or identification of posterior confidence with mean-sign accuracy is assumed. The fixed-margin Gaussian case $\gamma = 0$ and the positive subcritical regime are not covered by this theorem.

1 Model and main result

Let $\mathcal{X} = \{-1, 1\}^d$ with the uniform probability measure, and let $f(x) = \prod_{j=1}^d x_j$ be full parity. Although parity has Boolean circuits of size $O(d)$, its learnability depends on the specified prior and prediction rule. We analyze Gaussian-process classification with covariance

$$K(x, y) = H^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d + 1}\right), \qquad H(t) = \frac{2}{\pi} \arcsin t, \tag{1}$$

where the depth $L \ge 2$ is fixed. For a centered Gaussian vector $G \sim N(0, K)$ and training inputs $X_1, \dots, X_n$, the posterior conditions on the exact signs $f(X_i)G(X_i) > 0$. Its probability of being correct at x is

$$q_S(x) = \mathbb{P}\{f(x)G(x) > 0 \mid f(X_i)G(X_i) > 0 \text{ for all } i\}.$$

We count an input as a failure when $q_S(x) \le 2/3$. Thus the criterion concerns the actual posterior probability and includes equality as a failure. It is not identified with the error of the sign of a posterior mean.

Our main theorem, Theorem 6.2, proves that for some $c'_L > 0$ and every sufficiently large d, every integer sample size $0 \le n \le e^{c'_L d}$ satisfies

$$\mathbb{P}_S\{\mathbb{P}_x\{q_S(x) \le 13/20\} \ge \varepsilon\} \ge \frac{3}{4}, \qquad \varepsilon = \frac{2\pi(1/e - 7/20)^2}{256} > 7 \cdot 10^{-6}.$$

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The constants are deliberately conservative. The result gives a fixed positive failure fraction and does not claim a numerical fraction such as $1/10$. The threshold $13/20 < 2/3$ supplies slack for transferring the result to finite-width posterior probabilities.

The probability statement applies to each specified sample size; it does not assert that one event controls all prefixes of a common infinite sample simultaneously. The proof covers the entire interval of sample sizes directly and never assumes that this learner's fixed-target risk or confidence is monotone in the number of observations.

1.1 The finite-width network and its Gaussian limit

The same theorem applies to the following network convention. All hidden layers have width N, with

$$h^0(x) = (1, x), \quad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \le s < L), \quad z_N(x) = W^L h^{L-1}(x).$$

There are no additional biases. The weights are independent centered Gaussians with variances

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \quad \mathrm{Var}(W^s_{ij}) = N^{-\gamma} \ (1 < s < L), \quad \mathrm{Var}(W^L_i) = N^{-\gamma-1}. \tag{2}$$

Equivalently, prescribe the normalized weight scale $t(N) = N^{(1-\gamma)/2}$ before varying width. The three variances are $t(N)^2/d$, $t(N)^2/N$, and $t(N)^2/N^2$, respectively. Thus γ is fixed while the scalar scale follows this width schedule.

Before training, condition on the global margin event $A_a = \{|z_N(x)| > a \text{ for every } x \in \mathcal{X}\}$, where $a > 0$ is fixed. Training then conditions on $\operatorname{sign} z_N(X_i) = f(X_i)$. Let $q_{N,S}(x)$ be the resulting correct-label probability and $F_N(S) = \mathbb{P}_x\{q_{N,S}(x) \le 2/3\}$.

Proposition 1.1 (Conditional width limit)**.** *At fixed d, L, a, S and $\gamma < 0$, for every $x \in \mathcal{X}$,*

$$q_{N,S}(x) \longrightarrow q_S(x).$$

All these posteriors are well defined for sufficiently large N. Consequently

$$\liminf_{N\to\infty} F_N(S) \ge \mathbb{P}_x\{q_S(x) \le 13/20\}.$$

Proof. Let $Q^s_N(x, y) = N^{-1}\sum_i h^s_i(x)h^s_i(y)$. Conditional on the preceding layer, hidden rows are independent bounded random vectors. The conditional law of large numbers therefore applies to all empirical second moments, since the cube is finite. Preactivation variances grow like $N^{1-\gamma}$. Thus tanh converges to the sign function under a Gaussian coupling. The identity $\mathbb{E}[\operatorname{sign} U \operatorname{sign} V] = (2/\pi)\arcsin\rho$ for standard jointly Gaussian U, V of correlation ρ follows, for example, by measuring the corresponding angular sectors of a rotationally invariant planar Gaussian. Dominated convergence and induction over layers yield

$$Q^{L-1}_N \longrightarrow K \quad \text{in probability.}$$

For completeness, the convergence of the conditional moment map is uniform near its limiting covariance: if it were not, choose a converging covariance subsequence, couple by continuous matrix square roots, and apply dominated convergence. Its diagonal is positive, so the limiting scalar Gaussian coordinates vanish with probability zero. The first layer normalizes $(d+1)/d$ on the diagonal and hence gives the argument $(1 + x \cdot y)/(d+1)$ in (1).

Conditionally on the hidden layers, $N^{\gamma/2} z_N \sim N(0, Q^{L-1}_N)$. Therefore the whole output vector $G_N = N^{\gamma/2} z_N$ converges in law to $G \sim N(0, K)$. The global margin in these coordinates is

$|G_N(x)| > aN^{\gamma/2}$, whose threshold tends to zero. The spectral proof below shows K is positive definite. Hence all coordinate hyperplanes are Gaussian null sets and every sign orthant has positive probability. A shrinking-neighborhood argument around the coordinate hyperplanes shows that imposing this vanishing margin changes the probability of each fixed sign event by $o(1)$. Ratios of probabilities of sign events now give $q_{N,S}(x) \to q_S(x)$ and positivity of the conditioning probability for large N.

At a point where $q_S(x) \le 13/20$, convergence implies $q_{N,S}(x) < 2/3$ eventually. The cube is finite, so the implication holds simultaneously at all these points, proving the final assertion. □

The width limit is always taken at fixed d, L, γ, a, S. The subsequent exponential statement concerns $d \to \infty$ for these limiting learners. No uniform interchange of the width and dimension limits is asserted.

Proof roadmap and relation to later results. The next two sections establish Fourier estimates and a random tail-covariance bound. Gaussian sign conditioning then supplies a norm bound for the posterior mean and a latent margin at confident inputs. The final section inserts a thin middle band of Fourier degrees to cover every sample size, including gaps between successive level dimensions. Its small-sample argument is deterministic in the training locations. The later paper [1] includes this negative-exponent result in a theorem covering all $\gamma < 1$.

2 Fourier level masses of the kernel

Put $D = d + 1$, $\beta = 2^{-(L-1)}$, and $F = H^{\circ(L-1)}$. The characters $\chi_U(x) = \prod_{j\in U} x_j$ diagonalize the normalized covariance operator $M^{-1}K$, $M = 2^d$. Let λ_j be its eigenvalue at degree j. We first work on the full D-cube with kernel $F(u \cdot v/D)$, whose degree-j eigenvalue is denoted by a_j. Constants in this and the next section may depend on L. Notation $b \asymp c$ means two inequalities by positive constant multiples, uniformly in the stated ranges.

Lemma 2.1 (Power coefficients and paired eigenvalues)**.** *Write $F(z) = \sum_{m \text{ odd}} c_m z^m$. Every odd coefficient is positive, $\sum_m c_m = 1$, and*

$$c_m \asymp \frac{\Gamma(m-\beta)}{m!} \asymp m^{-1-\beta} \qquad (m \ge 1 \text{ odd}).$$

The full-cube eigenvalues vanish at even degrees and strictly decrease through odd degrees. They are positive at every odd degree. On the original d-cube,

$$\lambda_{2k} = \lambda_{2k+1} = a_{2k+1} > 0,$$

with a final unpaired eigenvalue if d is even. In particular K is positive definite.

Proof. The series of H has positive coefficients at every odd exponent, with

$$[z^{2k+1}]H(z) = \frac{2}{\pi}\frac{\binom{2k}{k}}{4^k(2k+1)}.$$

Composition preserves strict positivity and oddness. Near $z = 1$ in a slit neighborhood,

$$H(z) = 1 - b(1-z)^{1/2} + O((1-z)^{3/2}), \qquad b = \frac{2\sqrt{2}}{\pi}.$$

Induction gives $F(z) = 1 - A(1-z)^\beta + O((1-z)^{3\beta})$, with $A > 0$; at each composition its leading constant changes from A_t to $b\sqrt{A_t}$, starting from $A_1 = b$. These expansions admit algebraic coefficient transfer [2, Chapter VI]. Here are the analytic-domain details. On the unit circle outside $\{-1, 1\}$, positivity of every odd coefficient gives $|H(z)| < 1$: equality in the triangle inequality would require $z^2 = 1$. Hence the iterated functions extend analytically through every compact arc away from these endpoints. At 1, restrict $\arg(1-z)$ to a closed proper subinterval of $(-\pi, \pi)$. Each square-root composition reduces this angular range, and its positive leading coefficient keeps its image in the slit neighborhood for the next arcsine composition. Oddness gives the identical argument at -1. These local sectors and the compact arcs provide the required two-singularity indented domain. The contributions of 1 and -1 add at odd exponents, giving

$$c_m \sim \frac{2A}{-\Gamma(-\beta)} m^{-1-\beta} \quad (m \to \infty \text{ odd}).$$

Together with positivity of the finitely many initial coefficients and the gamma-ratio asymptotic, this proves the coefficient comparisons. For $L = 2$ the explicit formula and Stirling's bounds suffice instead.

Let $p_m(j)$ be the probability that the simple coordinate-flip walk on the D-cube reaches a specified distance-j vertex in m steps. Expanding $(u \cdot v/D)^m$ shows that its degree-j Fourier coefficient is $p_m(j)$. Thus $a_j = \sum_{m \text{ odd}} c_m p_m(j)$. For the walk recursion,

$$\begin{aligned} D[p_{m+1}(j) - p_{m+1}(j+2)] = j[p_m(j-1) - p_m(j+1)] \\ + (D-j-2)[p_m(j+1) - p_m(j+3)]. \end{aligned}$$

Induction from time zero gives nonincreasing probabilities through each parity class; terms with zero boundary coefficients are omitted. The term $m = j$ makes the odd-eigenvalue inequalities strict and proves positivity. Even-degree eigenvalues vanish by parity. Restricting the distinguished coordinate to $+1$ gives $\lambda_q = a_q + a_{q+1}$, proving the claims. □

For odd r, the exact combined multiplicity of reduced degrees $j-1, j$ is $\binom{d}{j-1} + \binom{d}{j} = \binom{D}{j}$. Consequently

$$D_r = \sum_{\substack{j \le r \\ j \text{ odd}}} \binom{D}{j}, \quad \tau_r = \sum_{\substack{j > r \\ j \text{ odd}}} \binom{D}{j} a_j, \quad \lambda_{r+1} = a_{r+2}. \tag{3}$$

The odd cutoff is essential for these exact identities.

Lemma 2.2 (Sharp level and tail estimates)**.** *Uniformly for odd $1 \le j \le D/4$ and odd $1 \le r \le D/8$,*

$$\binom{D}{j} a_j \asymp j^{-1-\beta}, \tag{4}$$

$$\tau_r \asymp r^{-\beta}, \tag{5}$$

$$\frac{\lambda_{r+1}}{\tau_r} \asymp \frac{1}{r\binom{D}{r+2}}, \tag{6}$$

$$s_{2,r} := \sum_{q > r} \binom{d}{q} \lambda_q^2 \le \frac{C r^{-2\beta-2}}{\binom{D}{r+2}}. \tag{7}$$

The same level-mass estimate holds up to $j \le \rho D$ for any fixed $\rho < 1/2$, with constants depending on ρ.

Proof. The gamma integral and Lemma 2.1 give, for odd j,

$$a_j \asymp D^{-\beta} \int_0^\infty u^{-1-\beta} p(u)^j (1-p(u))^{D-j}\, du, \quad p(u) = \frac{1-e^{-2u}}{2}, \tag{8}$$

$$\asymp D^{-\beta} \int_0^{1/2} p^j (1-p)^{D-j} w_\beta(p)\, dp, \tag{9}$$

where $w_\beta(p) = [-\frac{1}{2}\log(1-2p)]^{-1-\beta}/(1-2p)$. Indeed the gamma integral Poissonizes the walk time, and independent Poisson coordinate counts give independent coordinate parities with success probability $p(u)$. Since an odd-distance vertex has zero probability at even walk times, no extra parity factor occurs. The weight w_β is integrable near $1/2$ and comparable to $p^{-1-\beta}$ on $[0, b]$ for every fixed $b < 1/2$.

Fix $\rho < b < 1/2$ and take $j \le \rho D$. The integral on $[0, b]$ is comparable to the complete beta integral $B(j-\beta, D-j+1)$. For the lower comparison, a beta random variable with these parameters has mean at most ρ, so Markov's inequality puts at least a fraction $1-\rho/b$ of its mass in $[0, b]$. The identity

$$\binom{D}{j} B(j-\beta, D-j+1) = \frac{\Gamma(D+1)\Gamma(j-\beta)}{\Gamma(j+1)\Gamma(D+1-\beta)} \asymp D^\beta j^{-1-\beta}$$

gives the desired mass. On $[b, 1/2]$, the binomial factor decreases with p and, after multiplication by $\binom{D}{j}$, is at most $\exp[-D\,\mathrm{KL}(\rho\|b)]$. The remaining integrable weight makes this contribution exponentially small. This proves (4) and its stated extended range.

Sum (8) over the odd distances above r. Dropping the parity restriction and using the binomial Markov bound gives

$$\tau_r \le C D^{-\beta} \int_0^\infty u^{-1-\beta} \min\{1, Du/(r+1)\}\, du \le C(r+1)^{-\beta}.$$

For the lower bound, sum (4) over the odd j in $(r, 2r+2]$, using a fixed cutoff strictly below $D/2$ as needed at the endpoint. This yields $cr^{-\beta}$; bounded r are absorbed by a change of constants. Equation (6) follows from (4), (5), and (3).

For the second moment, use paired multiplicities to write $s_{2,r} = \sum_{j>r,\ j \text{ odd}} \binom{D}{j} a_j^2$. Up to an odd cutoff b_D near $D/4$, its terms are at most $Cj^{-2\beta-2}/\binom{D}{j}$. Consecutive odd inverse binomial coefficients have ratio

$$\frac{(j+1)(j+2)}{(D-j)(D-j-1)},$$

uniformly less than a fixed constant below one in this range. The sum is therefore bounded by a constant times its first term, at $j = r+2$. Beyond b_D, monotonicity gives the bound $a_{b_D}\tau_{b_D} \le CD^{-2\beta-1}/\binom{D}{b_D}$, which is exponentially smaller than the claimed bound uniformly for $r \le D/8$. This proves (7). □

Finally, two-step binomial ratios give, uniformly for odd $r \le D/8$,

$$\binom{D}{r} \le D_r \le C\binom{D}{r}, \qquad \frac{D_r}{\binom{D}{r+2}} \le C(r/D)^2. \tag{10}$$

All constants used here can be fixed independently of a subsequently chosen small positive limit $r/d \to \alpha$.

3 Random tail covariance matrices

Let Q be the covariance kernel retaining only degrees greater than an odd cutoff r, and write $\tau = \tau_r$, $\ell = \lambda_{r+1}$, and $s_2 = s_{2,r}$. This section proves the sample event needed later, without assuming that a Gaussian approximation remains valid after conditioning. The entire argument concerns the exact limiting Gaussian covariance.

Lemma 3.1 (Far-pair decay)**.** *There are constants $c_0, C > 0$ such that, when $r \le D/8$ is odd,*

$$\max_{d/4 \le \mathrm{dist}(x,y) \le 3d/4} |Q(x,y)| \le \rho_r := C\sqrt{D}\, e^{-r/8} + Ce^{-c_0 D}.$$

Proof. The full-cube Krawtchouk polynomial satisfies

$$\sum_{j=0}^{D} \mathsf{K}_j(h;D) z^j = (1+z)^{D-h}(1-z)^h.$$

If $\delta D \le h \le (1-\delta)D$, put $b = \min(h, D-h)$. On $|z| = t$ the generating function is bounded in absolute value by

$$(1+t^2)^b (1+t)^{D-2b} \le (1+t)^D \exp\left[-\frac{2\delta D t}{(1+t)^2}\right].$$

Cauchy's coefficient bound at $t = j/(D-j)$ and the uniform Stirling lower bound for $\binom{D}{j}$ therefore give, for $1 \le j < D$,

$$\frac{|\mathsf{K}_j(h;D)|}{\binom{D}{j}} \le C\sqrt{D}\exp[-2\delta j(1-j/D)]. \tag{11}$$

For reduced-cube points at distance in $[d/4, 3d/4]$, their augmented full-cube distance lies in $[D/8, 7D/8]$ for large d. Because r is odd, the exact tail is $Q(x,y) = \sum_{j>r,\ j \text{ odd}} a_j \mathsf{K}_j(h;D)$. For $r < j \le D - r$, (11) gives a bound $C\sqrt{D}e^{-r/8}$ after summation, since the total spectral mass is one. For $j > D - r$, monotonicity and an odd cutoff b_D near $D/4$ give

$$\sum_{j>D-r} \binom{D}{j} a_j \le a_{b_D} \sum_{k<r} \binom{D}{k} \le CD^{-\beta-1/2}\exp[-D(h(1/4) - h(1/8)) + O(1)].$$

The positive entropy gap absorbs the polynomial factor and proves the claim. Adjust constants for bounded d. □

Lemma 3.2 (A tail Gram deviation bound)**.** *Let $X_1, \dots, X_n$ be independent uniform cube points, with $1 \ll n \le M/4$. Let*

$$B_{\mathrm{near}} = \#\{y : \mathrm{dist}(x,y) \notin [d/4, 3d/4]\} \le 2e^{dh(1/4)}, \qquad q = 2\log M.$$

The count is independent of x. For every $z > 0$,

$$\begin{aligned}&\mathbb{P}\{\|Q_{SS} - \tau I_n\|_{\mathrm{op}} > z\tau\} \\ &\quad \le \frac{C}{z^2}\left[\frac{q\rho_r}{\tau} + \frac{\sqrt{nqs_2}}{\tau} + \frac{n\ell}{\tau} + \frac{n(1+B_{\mathrm{near}})}{M}\right]^2 + \frac{n^2(1+B_{\mathrm{near}})}{M}.\end{aligned} \tag{12}$$

Here the sampled matrix includes repeated training locations.

Proof. Let $J = Q - \tau I_M$, a zero-diagonal Hermitian matrix. Form a deterministic mask J_{far} retaining only pairs at distances in $[d/4, 3d/4]$. Since Q is positive semidefinite with constant diagonal τ, every entry has absolute value at most τ. Thus

$$\begin{aligned} \|J_{\mathrm{far}}\|_{\max} &\le \rho_r, \\ \|J_{\mathrm{far}}\|_{1\to 2}^2 &\le M s_2, \\ \|J_{\mathrm{far}}\|_{\mathrm{op}} &\le M\ell + \tau + \tau B_{\mathrm{near}}. \end{aligned}$$

For the second inequality, Parseval gives squared column norm $Ms_2 - \tau^2$ for J, and entrywise masking only decreases it. For the third, $\|J\| \le M\ell + \tau$, while the removed matrix has maximum absolute row sum at most τB_{near}. The entropy bound on B_{near} is the elementary binomial tail bound on distances to a fixed cube point.

Let R retain each of the M coordinates independently with probability $p = 2n/M$. The zero-diagonal case of Tropp's random principal-submatrix theorem [4, Theorem 1.1], at moment order two and $q = 2\log M$, states

$$(\mathbb{E}\|RJR\|^2)^{1/2} \le C\left[q(\mathbb{E}\|RJR\|_{\max}^2)^{1/2} + \sqrt{pq}(\mathbb{E}\|JR\|_{1\to 2}^2)^{1/2} + p\|J\|\right].$$

Apply it to J_{far} and substitute the preceding bounds:

$$(\mathbb{E}\|RJ_{\mathrm{far}}R\|^2)^{1/2} \le C[q\rho_r + \sqrt{nqs_2} + n\ell + (n/M)\tau(1 + B_{\mathrm{near}})]. \tag{13}$$

The logarithm here is that of the original dimension M, not the sample size.

To pass to a uniform subset of size n, use a uniform permutation of all cube points, independently of $B \sim \mathrm{Bin}(M, 2n/M)$. Its first B elements form the Bernoulli subset. On $B \ge n$, the first n principal submatrix is a compression of the larger one, so its operator norm cannot be larger. The size-n event is independent of B, and $\mathbb{P}(B \ge n) \ge 1/2$ for large n. Thus Markov's inequality applied to (13) gives the same fixed-size probability bound with a constant factor loss.

Conditional on no collisions, an independent size-n sample is a uniform size-n subset, up to an irrelevant ordering. Collisions have probability at most $n(n-1)/(2M)$. The probability of any sampled pair violating the distance condition is at most $n^2 B_{\mathrm{near}}/M$. On their complement, the sampled matrices J and J_{far} agree. Adding these exceptional probabilities proves (12). □

4 The actual Gaussian sign posterior

Let $\mathcal{X} = \{-1, 1\}^d$, $M = 2^d$, and $f(x) = \prod_{j=1}^d x_j$. All function norms and expectations on $\mathcal{X}$ use its uniform probability measure. Let G be a nondegenerate centered Gaussian vector indexed by $\mathcal{X}$, with covariance matrix K. The training inputs $X_1, \dots, X_n$ are independent and uniform. We condition on

$$\mathcal{A}_S = \{f(X_i)G(X_i) > 0 : 1 \le i \le n\}.$$

The prediction at x is its actual conditional correct-label probability $q_S(x) = \mathbb{P}(f(x)G(x) > 0 \mid \mathcal{A}_S)$, and we write

$$F_\theta(S) = \mathbb{P}_x\{q_S(x) \le \theta\}, \qquad \theta = \frac{13}{20}, \qquad c_\theta = \sqrt{2\pi}\left(\frac{1}{e} - \frac{7}{20}\right) > 0.$$

The threshold $\theta < 2/3$ supplies a strict buffer for passage from limiting Gaussian posteriors to finite-width posterior probabilities.

Suppose the Walsh characters are eigenfunctions of the normalized covariance operator, with degree-j eigenvalue λ_j. Write $V_r = \operatorname{span}\{\chi_U : |U| \le r\}$ and $D_r = \dim V_r$. For $r < d$, let Q be the covariance contributed by the complementary Fourier modes, and put

$$\tau = Q(x,x) = \sum_{j>r} \binom{d}{j} \lambda_j, \qquad \ell = \max_{j>r} \lambda_j.$$

The matrix eigenvalues of Q are $M\lambda_j$; this normalization is used throughout. Define $\|h\|_K^2 = h^T K^{-1} h$, and write $u_+ = \max(u, 0)$ and $u_- = \max(-u, 0)$.

Lemma 4.1 (Mean norm and a confidence margin)**.** *Let $m(x) = \mathbb{E}[G(x) \mid \mathcal{A}_S]$, and let $s \le n$ be the number of distinct sampled inputs. Then*

$$\|m\|_K^2 \le s, \qquad f(X_i)m(X_i) > 0.$$

Let $\sigma_x^2 = \operatorname{Var}(G(x) \mid G_S)$, where S here lists the distinct inputs. Whenever $\sigma_x > 0$ and $q_S(x) > \theta$,

$$f(x)m(x) > c_\theta \sigma_x.$$

Proof. Whiten the Gaussian training vector as $G_S = K_{SS}^{1/2}\xi$. The sign constraints form a homogeneous cone in ξ. Conditioning leaves the squared Gaussian radius with mean s, so Jensen's inequality and Gaussian regression give

$$m_S^T K_{SS}^{-1} m_S \le s, \qquad m = K_{\mathcal{X},S} K_{SS}^{-1} m_S.$$

The RKHS norm of this regression vector is the preceding quadratic form. Its observed signed coordinates are strictly positive.

For the confidence claim, the scalar conditional variable $Y = f(x)G(x)$ has a log-concave density: it is a marginal of a Gaussian restricted to a convex set, by the marginal log-concavity theorem [5]. If $\mu = \mathbb{E}Y$ and F_Y is its cdf, log-concavity and Jensen's inequality imply

$$\log F_Y(\mu) \ge \mathbb{E} \log F_Y(Y) = -1.$$

Consequently $\mathbb{P}(Y > \mu) \le 1 - e^{-1} < \theta$. Conditioning first on the full training vector shows that the density of Y is a mixture of Gaussian densities with common variance σ_x^2, and is therefore at most $(\sqrt{2\pi}\sigma_x)^{-1}$. If $q_S(x) > \theta$, then $\mu > 0$ and

$$\theta < \mathbb{P}(Y > 0) \le 1 - e^{-1} + \frac{\mu}{\sqrt{2\pi}\sigma_x}.$$

This gives the stated margin. □

Lemma 4.2 (Most residual variances remain large)**.** *For every training sample,*

$$\mathbb{P}_x\{\sigma_x^2 < \tau/2\} \le \frac{2n\ell}{\tau}.$$

Also, if p is the degree-r Fourier truncation of m and $t = m - p$, then $\|t\|_2^2 \le n\ell$.

Proof. Write $G = G_{\le r} + G_{>r}$ as a sum of independent fields. Conditioning additionally on the full low-degree field can only decrease conditional Gaussian variance. The remaining covariance is

$$Q - Q_{\mathcal{X},S} Q_{SS}^{\dagger} Q_{S,\mathcal{X}}.$$

Its subtracted term is positive semidefinite, bounded above by Q, and has rank at most $s \le n$. Its normalized trace is at most $n\ell$. Markov's inequality proves the first assertion; it includes observed points automatically. Fourier orthogonality and the mean norm bound prove $\|t\|_2^2 \le \ell \|m\|_K^2 \le n\ell$. □

5 Uniform sampling detects the negative part

Lemma 5.1 (Uniform ReLU sampling on the cube)**.** *For every fixed $\delta, \eta \in (0,1)$, there is a universal constant C such that, if*

$$n \geq C\eta^{-2} D_r \log\left(\frac{D_r}{\delta\eta}\right),$$

then with probability at least $1-\delta$, simultaneously for all $p \in V_r$,

$$\left|\frac{1}{n}\sum_{i=1}^{n}(-f(X_i)p(X_i))_+ - \mathbb{E}_x(-f(x)p(x))_+\right| \leq \eta\mathbb{E}_x|p(x)|.$$

In particular, when $r < d$ and $\eta = 1/4$, the empirical negative part is at least $\mathbb{E}|p|/4$.

Proof. Let A have rows $f(x)(\chi_U(x))_{|U|\leq r}$, one for each cube point. Then $A^TA = MI$ and every row has squared norm D_r. The ℓ_1 Lewis weights of A are exactly D_r/M: substitution in their defining equation

$$w_x^2 = A_x(A^T \operatorname{diag}(w^{-1})A)^{-1}A_x^T$$

verifies this constant solution, and Lewis weights are unique. The additive ReLU coreset theorem of Mai, Musco, and Rao [3, Theorem 5] therefore samples precisely the uniform distribution on the cube and gives the stated estimate. Its sample bound is $O(\eta^{-2}D_r\log(D_r/(\delta\eta)))$; no additional logarithmic factor is required. Finally $\mathbb{E}fp = 0$, because $r < d$, so $\mathbb{E}(-fp)_+ = \mathbb{E}|p|/2$. □

6 A uniform bound through the gaps between Fourier levels

We prove a lower bound for every sample size using a thin middle band of Fourier modes whose covariance trace is small. Ordinary Lipschitz contraction controls this band, while Lewis-weight sampling controls the lower degrees.

Lemma 6.1 (A fixed-threshold sampling bound)**.** *Let $r < u < d$, and put $\Delta = \tau_r - \tau_u$. Let $\mathcal{B}_{r,u}$ be the span of Walsh characters of degrees $r+1$ through u. Suppose*

$$\eta^2 \geq C\frac{D_r\log(2n)}{n}$$

for a sufficiently large universal constant C. For deterministic $R, T \geq 0$, the class

$$\mathcal{H} = \{-f(p+b) : p \in V_r,\ \mathbb{E}|p| \leq R,\ b \in \mathcal{B}_{r,u},\ \|b\|_K \leq \sqrt{n}\}$$

satisfies

$$\mathbb{E}_S \sup_{h\in\mathcal{H}} |\mathbb{P}_n(h-T)_+ - \mathbb{E}_x(h-T)_+| \leq 4(R\eta + \sqrt{\Delta}). \tag{14}$$

Here $\mathbb{P}_n g = n^{-1}\sum_i g(X_i)$.

Proof. For the full-cube matrix with rows $f(x)(\chi_U(x))_{|U|\leq r}$, the Lewis weights are D_r/M. The linear-process moment bound of Mai, Musco, and Rao [3, Theorem 3], with its failure parameter set to $1/2$, therefore implies

$$\mathbb{E}_{S,\epsilon} \sup_{p\in V_r,\,\mathbb{E}|p|\leq R}\left|\frac{1}{n}\sum_i \epsilon_i f(X_i)p(X_i)\right| \leq R\eta.$$

The normalization follows directly: their sampled matrix rescales each sampled row by M/n, whereas $\|p\|_1 = M^{-1}\|A\beta\|_1$. Their moment bound implies the displayed expectation bound by Jensen.

The covariance-weighted feature vector of the band has squared norm Δ at every input. Consequently

$$\mathbb{E}_{S,\epsilon} \sup_{\|b\|_K \le \sqrt{n}} \left| \frac{1}{n} \sum_i \epsilon_i f(X_i) b(X_i) \right| \le \frac{\sqrt{n}}{n} \sqrt{n\Delta} = \sqrt{\Delta}.$$

The map $z \mapsto (z - T)_+$ is 1-Lipschitz and vanishes at zero. Symmetrization, followed by the scalar contraction principle, bounds the left-hand side of (14) by four times the linear Rademacher width of $\mathcal{H}$. The triangle inequality and the preceding two estimates prove the claim. Both parameter balls and the threshold are fixed before sampling; no positive homogeneity in T or relative bound on an unbounded parameter class is asserted. □

Theorem 6.2 (Every sample size up to an exponential threshold)**.** *Fix $L \ge 2$. There exist $c'_L > 0$, $\varepsilon > 0$, and d_0 such that for every $d \ge d_0$ and every integer $0 \le n \le e^{c'_L d}$, independent uniform parity training inputs satisfy*

$$\mathbb{P}_S\{F_{13/20}(S) \ge \varepsilon\} \ge \frac{3}{4}.$$

The constants may be chosen with $\varepsilon = c_\theta^2/256$, where $c_\theta = \sqrt{2\pi}(1/e - 7/20)$. The same conclusion holds for $\liminf_{N\to\infty} F_N(S)$ at every fixed $\gamma < 0$ in the network convention of Proposition 1.1.

We first prove the range $e^{(\log(d+1))^3} \le n \le e^{c'_L d}$. A short argument for smaller samples follows the proof.

Proof in the larger-sample range. Fix a small sampling-failure probability $\delta = 1/20$. All constants below may depend on L and δ. Choose a sufficiently large constant A, to be fixed below, and let r be the largest odd integer such that

$$AD_r \log(D_r + 1) \le n. \tag{15}$$

By choosing c'_L sufficiently small, for all large d this cutoff satisfies

$$\tfrac{1}{4}(\log d)^2 \le r \le \alpha d, \qquad n < AD_{r+2} \log(D_{r+2} + 1),$$

where $\alpha > 0$ is a small constant chosen after A. Indeed $D_r \le (ed/r)^r$ gives the lower bound, while binomial entropy estimates give the upper bound. Uniformly in this range, $\log(2n) \le C \log(D_r+1) \le Cr \log(eD/r)$.

Put $\tau = \tau_r$, $\ell = \lambda_{r+1}$, $s_2 = s_{2,r}$, and

$$B = C_\delta \left(1 + \frac{\sqrt{nds_2}}{\tau} + \frac{n\ell}{\tau}\right), \qquad \eta^2 = C_0 \frac{D_r \log(2n)}{n}.$$

The constant C_0 is large enough for Lemma 6.1. The random Gram estimate implies $\|Q_{r,SS}\| \le B\tau$ with probability at least $1 - \delta$ for all large d. Its omitted far-pair terms tend to zero uniformly here: $r \ge (\log d)^2/4$, and c'_L is small enough that $2c'_L < \log 2 - h(1/4)$.

The crucial estimate is

$$\eta^2 B \le C_\delta \left[\frac{1}{A} + \sqrt{\frac{\rho \log(e/\rho)}{A}} + \rho^2 \log(e/\rho)\right], \qquad \rho = r/D. \tag{16}$$

To verify it, first $\eta^2 \le C/A$ by (15). Next, the sharp spectral estimates give

$$\eta^2 \frac{n\ell}{\tau} \le \frac{CD_r \log(2n)}{r\binom{D}{r+2}} \le C\rho^2 \log(e/\rho).$$

Finally, the square of the remaining term is at most

$$\left(\eta^2 \frac{\sqrt{nds_2}}{\tau}\right)^2 \le \frac{C}{A} \frac{CD_r \log(2n)\, d}{r^2\binom{D}{r+2}} \le \frac{C}{A}\rho \log(e/\rho).$$

This proves (16). Its right-hand side can be made arbitrarily small, uniformly over all these sample sizes, by choosing A large and then α small.

Let m be the actual sign-posterior mean. Its degree-r truncation p lies, on the Gram and ordinary ReLU sampling events, in the deterministic L^1 ball

$$\mathbb{E}|p| \le 4\sqrt{B\tau} =: R. \tag{17}$$

Indeed $\|m-p\|_K^2 \le n$ and the Gram event bound its empirical squared norm by $B\tau$. Training consistency and the ordinary ReLU estimate at accuracy $1/4$ then give (17), without a VC or squared-loss estimate. For sufficiently large A, that ReLU event has probability at least $1-\delta$.

Now choose the odd cutoff

$$u = r + 2\lceil 10 \log(d+1)\rceil,$$

and split $m = p + b + t$ into degrees at most r, degrees $r+1$ through u, and degrees greater than u. The sharp level masses give

$$\Delta := \tau_r - \tau_u \le C\tau_r \frac{\log d}{r} = o(\tau_r) \tag{18}$$

uniformly in the sample range. The upper shell threshold bounds $n\ell/\tau \le CA\log(eD/r)$ and $nds_2/\tau^2 \le CA(d/r)\log(eD/r)$. Below degree $D/8$, each successive odd binomial coefficient grows by at least a factor four. The additional $\lceil 10\log(d+1)\rceil$ steps therefore ensure

$$n\lambda_{u+1}/\tau = o(1), \qquad nds_{2,u}/\tau^2 = o(1), \qquad \tau_u/\tau = 1 - o(1).$$

Apply the random Gram estimate at u. With probability $1-o(1)$,

$$\mathbb{P}_n t^2 \le 2\tau, \qquad \|t\|_2^2 \le n\lambda_{u+1} = o(\tau). \tag{19}$$

The normalized trace of the tail covariance captured by training is at most $n\lambda_{u+1}$. The residual-variance argument thus also gives $\sigma_x^2 \ge \tau/2$ outside an $o(1)$ fraction of inputs. All these error terms are uniform in n.

Set $a_0 = c_\theta/(2\sqrt{2})$ and choose the deterministic threshold $T = 4\sqrt{\tau}/a_0$. Lemma 6.1, Markov's inequality, and (17) imply, with probability at least $1-\delta$, a uniform error at most

$$\frac{4}{\delta}(R\eta + \sqrt{\Delta})$$

for the empirical stoploss of every member of the indicated sum class. By (16) and (18), choose A large, α small, and then d large so that this is at most $a_0\sqrt{\tau}/8$. The actual $p+b$ belongs to this class on the events already specified, since $\|b\|_K \le \sqrt{n}$.

Write $h = p + b$ and $F = F_{13/20}(S)$. The confidence margin and (19) imply $fh > a_0\sqrt{\tau}$ on a set of measure at least $1 - F - o(1)$. Its negative set consequently has measure at most $F + o(1)$.

Moreover $\mathbb{E}fh = 0$ exactly, because $u < d$. Training consistency gives $(-f(X_i)h(X_i))_+ \le |t(X_i)|$, and $(z-T)_+ \le z^2/(4T)$ for $z \ge 0$. Hence

$$\mathbb{P}_n(-fh-T)_+ \le \frac{\mathbb{P}_n t^2}{4T} \le \frac{a_0\sqrt{\tau}}{8}.$$

The uniform stoploss estimate raises this bound to at most $a_0\sqrt{\tau}/4$ in the population. Equality of the positive and negative integrals of fh now yields

$$a_0\sqrt{\tau}(1-F-o(1)) \le \mathbb{E}(-fh)_+ \le T(F+o(1)) + \frac{a_0\sqrt{\tau}}{4}.$$

After division by $\sqrt{\tau}$, this implies $F \ge a_0^2/32 = c_\theta^2/256$ for all large d. The three events with failure probability δ, together with the event of failure probability $o(1)$, have joint probability at least $1 - 3\delta - o(1) > 3/4$. This completes the larger-sample proof. □

Lemma 6.3 (Parity obstruction for polynomial thresholds)**.** *Every polynomial threshold of degree at most $s < d$ has full-parity error at least*

$$B(d,s) = 2^{-d} \sum_{j=0}^{\lfloor (d-s-1)/2 \rfloor} \binom{d}{j}.$$

Proof. Use either fixed sign convention at zero. A sufficiently small constant perturbation then removes zero ties without changing the threshold's assigned signs. Let P be the resulting polynomial and $t = \lfloor (d-s-1)/2 \rfloor$. If the error set has fewer than $\dim V_t$ points, linear algebra supplies a nonzero $Q \in V_t$ vanishing at every error. Then fPQ^2 is everywhere nonnegative and strictly positive somewhere. But $\deg(PQ^2) \le s + 2t < d$, so its correlation with full parity is zero, a contradiction. □

Completion for smaller samples. For $n \le e^{(\log(d+1))^3}$ choose an odd $s \sim d^{1/3}$. The sharp spectral bound gives $n\lambda_{s+1}/\tau_s = o(1)$, uniformly in n. The posterior mean norm bound, residual-variance lemma, and confidence margin imply that its degree-s truncation agrees with parity on every confident input except an $o(1)$ fraction of inputs. Thus Lemma 6.3 gives

$$F_{13/20}(S) \ge B(d,s) - o(1) \ge \frac{1}{3}$$

for all large d, for every training sample. Here $B(d,s) \to 1/2$: by symmetry the removed central binomial window has $O(s)$ atoms, each of mass $O(d^{-1/2})$, and $s = o(\sqrt{d})$. This proves the remaining range. Proposition 1.1 transfers the confidence conclusion to every fixed negative exponent. □

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